



COMPETENCY STANDARD
FOR
3D FASHION DESIGN
(RMG SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for 3D Fashion Design (RMG sector) is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

This document has been developed to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)** to meet the skills requirements of the RMG sector. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (30 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-3FD-01-G	Apply occupational health and safety (OHS) practice in the workplace	1 Identify OHS policies and procedures. 2 Apply personal health and safety practices. 3 Report hazards and risks. 4 Respond to emergencies	08
SICIP-RMG-3FD-02-G	Manage personal and professional development	1 Interpret personal development skills. 2 Set and meet self-development priorities. 3 Maintain professional growth and development.	10
SICIP-RMG-3FD-03-G	Support innovation and manage change	1 Identify needs for innovation in the workplace 2 Apply creative approach and solution 3 Support flexible and innovative ways of working 4 Adapt to emerging technological changes and opportunities	12
Total Hour			30

Sector Specific Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-3FD-01-S	Apply Quality Procedures	1. Identify quality procedures 2. Follow quality procedure 3. Maintain standard procedure	10
SICIP-RMG-3FD-02-S	Apply green practices in RMG sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
Total Hour			20

Occupation Specific Competencies (222 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-RMG-3FD-01-O	Interpret fundamentals of 3D fashion design	1. Understand digital garment construction/style 2. Analyze garment structure 3. Interpret 2D to 3D garment visualization flow 4. Identify hardware and software used in 3D fashion design	20
SICIP-RMG-3FD-02-O	Operate 3D Fashion Design Software	1. Install and configure essential 3D fashion design software 2. Navigate user interface and workspace tools 3. Import digital patterns and 3D models 4. Customize avatars and fabric libraries 5. Perform real-time 3D simulations	100
SICIP-RMG-3FD-03-O	Design Digital Garments in 3D Environment	1. Create basic garments in 3D 2. Apply digital fabrics and textures 3. Adjust pattern fittings in a virtual environment 4. Design trims, accessories, and graphics placement 5. Export 3D garment prototypes	50
SICIP-RMG-3FD-04-O	Analyze and Modify 3D Garment Fit	1. Conduct virtual fit sessions using 3D avatars 2. Modify patterns based on fit analysis 3. Evaluate garment comfort and movement 4. Collaborate with technical design teams for refinement	32
SICIP-RMG-3FD-05-O	Collaborate in	1. Collaborate with teams	20

Code	Unit of Competency	Elements of Competency	Duration (hours)
	Digital Product Development Cycle	using 3D design tools 2. Incorporate feedback to optimize virtual samples 3. Optimize approval processes with 3D tools	
Total Hours			222

COMPETENCY STANDARD: 3D FASHION DESIGN

Generic Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE	Nominal Duration: 08 hrs.	Unit Code: SICIP-RMG-3FD-01-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the task of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are interpreted 1.2 Safety signs and symbols are identified and followed. 1.3 Response, evacuation procedures and other contingency measures are interpreted correctly.
2. Apply personal health and safety practices	2.1 OHS policies and procedures are interpreted in the workplace including <u>personal protective equipment (PPE)</u> . 2.2 Common health issues are recognized. 2.3 Common safety issues are identified.
3. Report hazards and risks	3.1 Hazards and risks are identified. 3.2 Hazards and risks assessment and controls are interpreted
4. Respond to emergencies	4.1 Respond to alarms and warning devices. 4.2 <u>Emergency response plans and procedures</u> are responded to. 4.3 <u>First aid procedures</u> during emergency situations are identified.

Range of variables:

Variables	Range (may include but not limited to)
1. OHS policies	1.1 Organizational OHS policies 1.2 International OHS requirements 1.3 Fire safety rules and regulations
2. Emergency response plans and procedures	1.1 Firefighting procedures 1.2 Earthquake response procedures

	1.3 Emergency response plans and procedures 1.4 Medical and first aid
3. First aid procedure	1.1 Washing of open wound 1.2 Washing chemically infected area 1.3 Applying bandage 1.4 Taking appropriate medicine
4. Personal protective equipment	1.1 Safety glasses 1.2 Ear plugs 1.3 Gloves 1.4 Apron 1.5 Helmet 1.6 Mask 1.7 Safety shoes

Curricular Content Guide

1. Underpinning knowledge	1.1 Workplace OHS policies and procedures 1.2 Work safety procedures 1.3 Emergency response procedures: 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 OHS awareness 1.6 Personal protective equipment (PPE) 1.7 Malfunctions and resolutions
2. Underpinning skill	2.1 Identifying OHS policies and procedures 2.2 Applying personal health and safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergencies
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource implications	The following resources must be provided: 3.1 Workplace (simulated or actual) 3.2 Personal protective equipment (PPE) 3.3 Firefighting equipment 3.4 Emergency response manual 3.5 First aid kits 3.6 Stationery 3.7 Learning manual.

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified OHS policies and procedures 1.2 applied personal health and safety practices 1.3 reported hazards and risks 1.4 responded to emergencies.
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: MANAGE PERSONAL AND PROFESSIONAL DEVELOPMENT	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-3FD-02-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to manage personal and professional development. It specifically includes the task of interpreting personal development skills, setting and meeting self-development priorities and maintaining professional growth and development.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret personal development skills	1.1 Objectives of personal development skills are described. 1.2 <u>Personal development skills</u> are identified. 1.3 Intra and Interpersonal relationships are maintained in the course of managing oneself. 1.4 Self-analysis is performed and personal development needs are identified.
2. Set and meet self-development priorities	1.5 Tasks are prioritized to achieve personal, team and organizational goals and objectives. 1.6 <u>Resources</u> are utilized efficiently and effectively to manage work priorities and commitments. 1.7 Economic usage and maintenance of facilities are followed as per established procedures.
3. Maintain professional growth and development	1.8 Pro activeness/zeal is demonstrated in fulfilling personal and professional growth requirements. 1.9 <u>Trainings and career opportunities</u> are identified and accessed based on job requirements. 1.10 <u>Recognitions</u> are sought/ received and demonstrated as proof of career advancement. 1.11 <u>Licenses and/or certifications</u> relevant to the job and career are obtained and renewed

Range of variables:

Variables	Range (may include but not limited to)
1. Personal Development skills	1.1 Problem-solving 1.2 Self-confidence 1.3 Adaptability 1.4 Integrity 1.5 Work ethic 1.6 Pro-activeness
2. Resources	1.7 Human

	1.8 Financial 1.9 Technology
3. Trainings and career opportunities	1.10 Participation in training programs 3.1.1 Technical 3.1.2 Supervisory 3.1.3 Managerial 3.1.4 Continuing Education 1.11 Serving as Resource Persons in conferences and workshops
4. Recognitions	1.12 Recommendations 1.13 Citations 1.14 Certificate of Appreciations 1.15 Commendations 1.16 Awards 1.17 Tangible and Intangible Rewards
5. Licenses and/or certifications	1.18 National Certificates 1.19 Certificate of Competency 1.20 Support Level Licenses 1.21 Professional Licenses

Curricular Content Guide

1. Underpinning knowledge	1.1 Importance of personal development skills. 1.2 Organizational policies relevant to training and professional growth. 1.3 Company operations, procedures and standards. 1.4 Resources in work environment.
2. Underpinning skill	1.5 Utilizing and improving personal development skills 1.6 Maintaining Intra and Interpersonal relationship 1.7 Utilizing communication skills 1.8 Prioritizing tasks in accordance with work commitment. 1.9 Utilizing resources efficiently and effectively
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 1.1 Workplace (simulated or actual) 1.2 Workplace procedures 1.3 Standard operating procedure 1.4 Workplace documents, signs and symbols 1.5 Codes of conduct

	1.6 Projector
	1.7 Stationery
	1.8 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 interpreted personal development skills. 1.2 set and meet self-development priorities. 1.3 maintained professional growth and development.
2. Methods of assessment	Competency should be assessed by: 1.4 Written test 1.5 Practical Demonstration 1.6 Oral Questioning 1.7 Portfolio (Optional)
3. Context of assessment	1.8 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 1.9 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: SUPPORT INNOVATION AND MANAGE CHANGE	Nominal Duration: 12 hrs.	Unit Code: SICIP-RMG-3FD-03-G
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to support innovation and manage change. It specifically includes the task of identifying needs for innovation in the workplace, applying creative approach and solution, supporting flexible and innovative ways of working and adapting to emerging changes and opportunities.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify needs for innovation in the workplace	1.1. The need for <u>innovation</u> in scope of work is recognized 1.2. The value of <u>innovative practices</u> in the workplace is recognized 1.3. Existing way of working in the workplace are interpreted. 1.4. Drawbacks of existing way of working are identified. 1.5. Benefits of change are identified to make it consistent with organizational requirements 1.6. Realistic timelines and targets for implementation of changes are set.
2. Apply creative approach and solution	2.1 Opportunities and creative approaches to implement innovation practices are identified 2.2 Creative approaches of coworkers pertaining to work practices are analyzed and incorporated 2.3 Innovation in accordance with organizational requirements changes are implemented
3. Support flexible and innovative ways of working	3.1 Maximum innovative opportunities are promoted. 3.2 Work assignments to facilitate innovative work skills are organized. 3.3 Team members are provided with guidance and coaching on innovation in the workplace. 3.4 Models of innovative work practice are provided and discussed 3.5 Appropriate environment for learning and innovation is maintained.
4. Adapt to emerging technological changes and opportunities	4.1 Usages of different <u>technologies</u> is determined based on job requirements. 4.2 Appropriate technology is selected as per work specifications. 4.3 Relevant technology is effectively used in carrying out functions.

	4.4 Appropriate implementation tools are used as per task requirement.
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Range of variables:

Variables	Range (may include but not limited to)
1. Innovative practices	1.1. Self-directed support to implement new ideas 1.2. Collaborative arrangement to apply new method of working 1.3. Making scope of work more efficient to use new technology
2. Innovation	1.4. New ideas 1.5. Different ideas 1.6. New methods of doing work 1.7. Use of new Technology
3. Technologies	1.8. Office technology 1.9. Industrial technology 1.10. System technology 1.11. Information technology 1.12. Training technology

Curricular Content Guide

1. Underpinning knowledge	1.1. Definition of innovation 1.2. Current practice in own scope of work 1.3. Workplace procedures 1.4. Support required to generate creative ideas 1.5. Difference between innovation and creativity 1.6. Common effects of change and innovation in the workplace 1.7. Industrial and organizational context of change 1.8. Organization's policies, plans, procedures and structure 1.9. Knowledge of resources required by the organization's operations 1.10. Awareness on technology and its function 1.11. Operating instructions 1.12. Applicable software 1.13. Company policy in relation to relevant technology 1.14. Technology adaptability
2. Underpinning skill	1.1. Identifying resources required for creativity and innovation 1.2. Contributing to brainstorming session 1.3. Identifying issues and concerns of one's scope of work 1.4. Encouraging co-workers to generate and develop ideas 1.5. Evaluating potential obstacles to and opportunities for

	creativity and innovation 1.6. Sharing of best practices related to innovation and creativity 1.7. Applying relevant technology 1.8. Using update machine and Software 1.9. Acquiring troubleshooting skills Identifying issues 1.10. Identifying current industry standard diagnostic tools 1.11. Describing common malfunctions and resolutions. 1.12. Determining the root cause of a routine malfunction
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical aspects of competency	Assessment required evidence that the candidate: 1.1 identified need for innovation in the workplace 1.2 applied creative approach and solution 1.3 supported flexible and innovative ways of working 1.4 adapted to emerging technological changes and opportunities
2. Methods of assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY QUALITY PROCEDURES	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-3FD-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to apply quality procedures. It specifically includes the task of identifying quality procedures, following quality procedure and maintaining standard procedure.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify quality procedures	1.1. <u>Manuals</u> and guidelines for RMG production processes are collected as per sample specifications. 1.2. The importance of manuals in ensuring quality standards is recognized. 1.3. Instructions and procedures for quality control are identified. 1.4. Required information is collected from manuals. 1.5. Performance measurement systems are identified.
2. Follow quality procedure	2.1 Instructions and procedures are followed strictly and duties are performed in accordance with demand of <u>quality improvement system</u> . 2.2 Concept of supplying product or service to meet the <u>customer quality requirements</u> is understood and accordingly applied. 2.3 Conformance to specifications is ensured. 2.4 Defects are detected and reported to the relevant authority according to standard operating procedures
3. Maintain standard procedure	3.1 Performance is assessed at regular interval. 3.2 Specifications and standard operating procedures are established 3.3 Quality of product is checked and verified. 3.4 Quality control and quality assurance system procedures for each job are followed. 3.5 Conformance to specification is ensured in every case at all situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. Manuals	1.1 Buyers' specification manual 1.2 Compliance manual 1.3 Maintenance procedure manual

	1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Signs and symbols, instruction manuals
2. quality improvement system.	2.1. Quality inspection 2.2. Testing 2.3. Quality control. 2.4. Quality assurance 2.5. Total Quality Management
3. customer quality requirements	3.1. Performance 3.2. Features 3.3. Reliability 3.4. Conformance 3.5. Aesthetics 3.6. Durability

Curricular Content Guide

1. Underpinning Knowledge	1.1 Importance of maintaining quality 1.2 quality, quality assurance, quality control, quality inspection, quality improvement and total quality control. 1.3 Process and procedures for improving and maintaining quality 1.4 Procedures for addressing defects. 1.5 Record keeping within the quality improvement system in workplace 1.6 Factors, which affect successful implementation of the quality systems and procedures.
2. Underpinning Skills	2.1. Maintaining good quality 2.2. Eliminating poor quality 2.3. Understanding the meaning of the key terms - quality, quality assurance, quality control, quality inspection, quality improvement and total quality control. 2.4. Improving and maintaining quality 2.5. Addressing defects and procedures 2.6. Recording within the quality improvement system in workplace. 2.7. Implementing quality systems and procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual)

	4.2 Map/globe 4.3 Projector 4.4 Stationery 4.5 Learning manual.
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified quality procedures 1.2 followed quality procedure 1.3 maintained standard procedure
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES IN RMG SECTOR	Nominal Duration: 10 hrs.	Unit Code: SICIP-RMG-3FD-02-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in RMG sector. It specifically includes the task of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices in RMG (Ready-Made Garment)</u> manufacturing are explained. 1.2 Key terms and symbols in RMG production processes are identified. 1.3 <u>Sources of environmental impacts</u> during RMG manufacturing activities are identified. 1.4 RMG manufacturing activities use are explained. 1.5 <u>Ways of mitigating environmental impacts</u> in RMG manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in RMG manufacturing are documented. 2.2 <u>Recyclable and non-recyclable items</u> in RMG production processes are identified. 2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented. 2.4 Sustainable alternatives in RMG manufacturing are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste in RMG manufacturing</u> are identified. 3.2 Hazardous waste in RMG production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in RMG manufacturing processes. 1.2 6R for waste management in RMG production: 1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste generation.

	1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in RMG manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in RMG production with low environmental impact. 1.4 Recycling of materials used in the RMG manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.
2. Sources of environmental impacts	2.1 Air Pollution in RMG Manufacturing 2.2 Water Pollution in RMG Manufacturing 2.3 Soil Degradation in RMG Manufacturing 2.4 Noise Pollution in RMG Manufacturing 2.5 Waste Generation in RMG Manufacturing 2.6 Resource Depletion in RMG Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in RMG production processes. 3.2 Adopting renewable energy sources for RMG manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 Recyclable Items in RMG Manufacturing 4.1.1 Metal components. 4.1.2 Wooden materials. 4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 Non-Recyclable Items in RMG Manufacturing 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.
5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products.

	5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.
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Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in RMG manufacturing 1.6 Hazardous waste in RMG production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Explaining principles of green practices in RMG manufacturing 2.2 Identifying sources of environmental impacts 2.3 Explaining re-cyclable and non-recyclable items 2.4 Minimizing resources use in the workplace 2.5 Identifying different types of waste in RMG manufacturing 2.6 Disposing hazardous waste in RMG production 2.7 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Projector 4.3 Stationery 4.4 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace
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	1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Occupation Specific Competencies

Unit of Competency: INTERPRET FUNDAMENTALS OF 3D FASHION DESIGN	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG-3FD-01-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to interpret fundamentals of 3D fashion design. It specifically includes the task of understanding digital garment construction/style, analyzing garment structure, interpreting 2D to 3D garment visualization flow and identifying hardware and software used in 3D fashion design.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand digital garment construction/style	1.1. Overview of garments construction/style is interpreted. 1.2. Spec sheet is identified as per job requirements. 1.3. <u>Relevant sewing methods/stitch types</u> are identified. 1.4. <u>Type of seams</u> are identified.
2. Analyze garment structure	2.1 <u>Garments structures</u> are identified as per job requirements. 2.2 Difference between <u>pattern components</u> and garment structure is analyzed. 2.3 Pattern components and garment structure is assessed and analyzed. 2.4 <u>Pattern markings</u> are identified. 2.5 Identification process of pattern markings is carried out.
3. Interpret 2D to 3D garment visualization flow	3.1 2D or flat pattern is interpreted. 3.2 2D pattern layout is identified. 3.3 2D pattern shape and lines are identified. 3.4 2D to 3D pattern placement procedure are interpreted. 3.5 <u>Assembling process</u> of sewing patterns are identified and interpreted. 3.6 <u>3D garments simulation process</u> is identified.
4. Identify hardware and software used in 3D fashion design	4.1 Hardware is identified in 3D fashion design. 4.2 Functions of hardware in 3D fashion design. 4.3 <u>3D Software</u> is identified in 3D fashion design. 4.4 Functions of different types of software is interpreted in 3D fashion design.

Range of Variables

Variable	Range (Includes but not limited to):
1. Relevant sewing methods/stitch type	1.1 Lock stitch 1.1.1 Single needle

	1.1.2 Double needle 1.1.3 Four needles 1.2 Chain stitch 1.2.1 Two threads 1.2.2 Three threads 1.2.3 Four threads 1.2.4 Five threads
2. Type of seams	2.1 Superimposed Seam 2.2 Lapped Seam 2.3 Bound Seam 2.4 Flat Seam 2.5 Decorative Seam 2.6 Edge Finishing Seam
3. Garments structures	3.1 Knit 3.2 Woven 3.3 Denim 3.4 Lingerie
4. Pattern components	4.1 Pattern components for tops 4.1.1 Front part 4.1.2 Back part 4.1.3 Sleeves 4.1.4 Collar 4.1.5 Cuff 4.1.6 Yoke 4.1.7 Placket 4.1.8 Hem Facing 4.1.9 Pocket 4.1.10 Neck Facing / Neckline Binding 4.2 Pattern components for bottoms 4.2.1 Front part 4.2.2 Back part 4.2.3 Waistband 4.2.4 Pocket Bags 4.2.5 Fly 4.2.6 Belt Loops 4.2.7 Yoke 4.2.8 Crotch Panel 4.2.9 Coin Pocket
5. Pattern markings	5.1 Notch 5.2 Dart 5.3 Placement marking
6. Assembling process	6.1 Free sewing 6.2 Segment sewing
7. 3D garments simulation process	7.1 Pattern Creation (2D CAD Pattern Design) 7.2 3D Garment Assembly 7.3 Fabric Simulation 7.4 Simulation & Animation

	7.5 Fitting & Alteration 7.6 Rendering & Presentation 7.7 Integration with Production
8. 3D software	8.1 BROWZWEAR-3D (VStitcher) 8.2 CLO-3D (Marvellous) 8.3 Style-3D 8.4 OPTITEX-3D 8.5 Lectra-3D 8.6 Garber-3D

Curricular Content Guide

1. Underpinning Knowledge	1.1 Document detailing design, fabric, trims, and measurements for garment production (Spec sheet) 1.2 Construction of garments, including fabric and stitching types 1.3 Techniques for assembling garments 1.4 Types of stitches (straight, zigzag, overlock) for different garment parts and finishes. 1.5 Internal and external garment design, including linings, trims, and construction features. 1.6 Basic garment pieces (front, back, sleeves, collar, waistband) cut from fabric. 1.7 Symbols on patterns indicating folds, sewing lines, and alignment. 1.8 Two-dimensional garment shape used as a cutting template for fabric. 1.9 Arrangement of pattern pieces on fabric to minimize waste. 1.10 Outline and measurements of pattern pieces for garment assembly. 1.11 Steps for stitching garment pieces, including machine setup and component attachment. 1.12 Using software to create a virtual 3D version of the garment for fabric behavior testing.
2. Underpinning Skills	2.1 Interpreting digital garment construction/style 2.2 Identifying garment's structure 2.3 Analyzing pattern components 2.4 Assessing garment's structure 2.5 Identifying 3D garments simulation process 2.6 Identifying hardware 2.7 Identifying 3D software
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness

	3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (actual or simulated) 4.2 Spec sheet 4.3 3D software 4.4 Learning manual 4.5 Accessibility to the workplace

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 understood digital garment construction/style 1.2 analyzed garment structure 1.3 interpreted 2D to 3D garment visualization flow 1.4 identifying hardware and software used in 3D fashion design
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: OPERATE 3D FASHION DESIGN SOFTWARE	Nominal Duration: 100 Hrs.	Unit Code: SICIP-RMG-3FD-02-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to operate 3D fashion design software. It specifically includes the task of installing and configuring essential 3D fashion design software, navigating user interface and workspace tools, importing digital patterns and 3D models, customizing avatars and fabric libraries and performing real-time 3D simulations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install and configure essential 3D fashion design software	1.1. 3D fashion software is identified in accordance with the job requirements. 1.2. 3D fashion software is installed. 1.3. 3D fashion software is configured. 1.4. 3D fashion software is checked and tested.
2. Navigate user interface and workspace tools	2.1. User interface is identified and interpreted. 2.2. <u>Workspace tools</u> are identified. 2.3. Appropriate user interface and workspace tools are navigated and applied.
3. Import digital patterns and 3D models	3.1 Digital pattern formats are identified as per the job. 3.2 File integrity and compatibility are verified before import operations are initiated. 3.3 <u>Software</u> is used for importing with default or custom import settings. 3.4 Pattern pieces are checked for missing data or formatting errors after import. 3.5 Imported patterns are aligned with digital mannequins or avatars using virtual fitting tools.
4. Customize avatars and fabric libraries	4.1. Importance of customizing avatars are interpreted. 4.2. Avatars are customized as per the requirements. 4.3. Digital fabric libraries are identified. 4.4. Digital fabric libraries are organized for testing and visualizing avatars. 4.5. All updates are documented and version-controlled to maintain data integrity.
5. Perform real-time 3D simulations	5.1 Appropriate simulation software and digital garment assets are selected. 5.2 Fabric and material properties are applied accurately to reflect realistic drape and behavior during simulation. 5.3 Real-time simulations are initiated and monitored to assess garment fit, movement, and strain mapping on 3D

	avatars. 5.4 <u>Software errors</u> are identified and corrected using in-software tools. 5.5 Simulated garments are adjusted iteratively. 5.6 Final simulations are rendered or exported. 5.7 Simulation settings are configured to test the garment under dynamic conditions.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Workspace tools	1.1 2D pattern tools 1.2 3D garments simulation tools 1.3 Sewing tools 1.4 Avatar tools 1.5 Fabric and materials tools 1.6 Rendering tools 1.7 Measurement tools 1.8 Grading tools
2. Software	2.1 Garber 2.2 Lectra 2.3 OPTITEX 2.4 Gemini
3. Software errors	3.1 Collision errors 3.2 Garment clipping 3.3 Mesh distortions

Curricular Content Guide

1. Underpinning Knowledge	1.1 Software used in a 3D environment. 1.2 Tools within the software used. 1.3 The layout of controls and navigation elements. 1.4 A virtual representation of garment patterns. 1.5 Ensuring the saved files maintain structure. 1.6 Virtual models used to simulate. 1.7 Customizing avatars to match specific body types. 1.8 A collection of virtual fabric types. 1.9 Understanding how different fabrics and materials 1.10 Software features that allow users to see immediate changes.
2. Underpinning Skills	2.1 Installing and configuring essential 3D fashion design software 2.2 Navigating user interface and workspace tools 2.3 Importing patterns 2.4 Organizing digital fabric libraries 2.5 Documenting all updates 2.6 Documenting version-control 2.7 Applying fabric properties and material 2.8 Adjusting simulating garments

	2.9 Rendering final simulations 2.10 Configuring simulation settings
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4 Resource Implications	4.1 Workplace (actual or simulated) 4.2 Computer/ Laptop 4.3 3D software tools 4.4 Accessibility to the workplace

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 installed and configured essential 3D fashion design software 1.2 navigated user interface and workspace tools 1.3 imported digital patterns and 3D models 1.4 customized avatars and fabric libraries 1.5 performed real-time 3D simulations
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: DESIGN DIGITAL GARMENTS IN 3D ENVIRONMENT	Nominal Duration: 50 Hrs.	Unit Code: SICIP-RMG-3FD-03-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to design digital garment in 3D environment. It specifically includes the task of creating basic garments in 3D, applying digital fabrics and textures, adjusting pattern fittings in a virtual environment, designing trims, accessories and graphics placement and exporting 3D garment prototypes.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Create basic garments in 3D	1.1 Basic 2D pattern blocks are imported into the 3D design software interface. 1.2 Pattern pieces are arranged and stitched virtually to construct the garment's structure on the 3D avatar. 1.3 Fabric properties and textures are applied to simulate realistic material behavior and appearance. 1.4 The garment is simulated and adjusted to ensure proper fit, alignment, and symmetry on the digital mannequin. 1.5 Basic design elements are incorporated to complete the garment construction. 1.6 Final garment is saved and documented.
2. Apply digital fabrics and textures	2.1. Digital fabric swatches are sourced or created using appropriate software tools. 2.2. Fabric properties are defined accurately within the 3D simulation environment. 2.3. Textures and prints are mapped into garment surfaces to ensure proper alignment and scaling. 2.4. Material shaders and lighting settings are configured to enhance fabric realism in both static and animated views. 2.5. Final textures and materials are reviewed and rendered.
3. Adjust pattern fittings in a virtual environment	3.1 Digital garment fittings are assessed using simulation tools. 3.2 Pattern pieces are resized based on avatar measurements. 3.3 Seam lines and darts are edited. 3.4 Strain maps and fit visualizations are interpreted.
4. Design trims, accessories, and graphics placement	4.1 Appropriate trims, accessories, and graphic elements are selected based on the design concept. 4.2 <u>Digital libraries</u> are utilized. 4.3 Placement of trims and accessories are executed. 4.4 <u>Graphics</u> are mapped as per job requirements. 4.5 <u>AI driven software</u> is used to generate graphics.

	4.6 Final placements are validated and documented for review.
5. Export 3D garment prototypes	5.1 The 3D garment prototype is finalized with accurate fit, textures, and simulation. 5.2 Appropriate file formats are selected based on the intended platform or end use. 5.3 Texture maps, material properties, and animation data are embedded or exported. 5.4 Export settings are configured correctly to preserve visual integrity. 5.5 Final export files are documented and submitted according to project or client specifications.

Range of Variables

Variable	Range (Includes but not limited to):
1. Digital libraries	1.1 Zippers 1.2 Buttons 1.3 Stitching details 1.4 Decorative elements 1.5 Stage 1.6 Hangers 1.7 Avatar
2. Graphics	2.1 Prints 2.1.1 Rubber print 2.1.2 Embossed print 2.1.3 Foil print 2.1.4 All over print 2.1.5 Flock print 2.1.6 Glitter print 2.2 Logos 2.2.1 Patch 2.2.2 Embroidery
3. File formats	3.1 JPG 3.2 JPEG 3.3 PDF 3.4 Turn table image 3.5 ai
4. AI driven software	4.1 ChatGPT AI 4.2 Leonardo AI 4.3 Midjourney AI

Curricular Content Guide

1. Underpinning Knowledge	1.1 Standardized flat pattern pieces 1.2 Characteristics of fabric 1.3 Fundamental components of design 1.4 Virtual representations of fabric samples
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	1.5 Key attributes of fabric 1.6 Textures and prints 1.7 Material shaders and lighting settings 1.8 Collections of pre-designed digital patterns, textures, and fabric swatches 1.9 Software that uses artificial intelligence
2. Underpinning Skills	2.1 Applying fabric properties and textures 2.2 Creating garments in 3D 2.3 Applying digital fabrics and textures 2.4 Adjusting pattern fittings in a virtual environment 2.5 Designing trims, accessories and graphics placement
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Environmental concerns 3.5. Eagerness to learn 3.6. Tidiness and timeliness 3.7. Respect for rights of peers and seniors in the workplace 3.8. Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (actual or simulated) 4.2 Computer/ laptop 4.3 3D software 4.4 Accessibility to the workplace

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 created basic garments in 3D 1.2 applied digital fabrics and textures 1.3 adjusted pattern fittings in a virtual environment 1.4 designed trims, accessories and graphics placement 1.5 exported 3D garment prototypes
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: ANALYZE AND MODIFY 3D GARMENT FIT	Nominal Duration: 32 Hrs.	Unit Code: SICIP-RMG-3FD-04-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to analyze and modify 3D garment fit. It specifically includes the task of conducting virtual fit sessions using 3D avatars, modifying patterns based on fit analysis, evaluating garment comfort and movement and collaborating with technical design teams for refinement.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Conduct virtual fit sessions using 3D avatars	1.1 Suitable 3D avatars are selected or customized. 1.2 Garments are draped on 3D avatars to assess fit, flow and overall design accuracy. 1.3 Simulation tools are used to assess fabric behavior. 1.4 <u>Fit issues</u> are identified by analyzing garment interactions. 1.5 Garments are simulated as per job requirements to ensure design specifications. 1.6 Feedback from virtual fit sessions is documented and applied to refine garment fit and design accuracy.
2. Modify patterns based on fit analysis	2.1 The fit analysis results are identified. 2.2 Pattern pieces are adjusted based on the fit issues. 2.3 Modifications to the pattern are made to correct issues in the garment's structure. 2.4 The altered patterns are re-simulated. 2.5 Feedback from the fit analysis is applied for enhanced garment performance. 2.6 The modified patterns are saved with proper version control for future reference and further development. 2.7 Final adjusted patterns are validated using both static and dynamic simulations.
3. Evaluate garment comfort and movement	3.1 The garment simulation is reviewed in both static and animated states for accuracy using updated software. 3.2 Avatar movements are animated to reflect garment interaction. 3.3 Fabric properties are reviewed and adjusted for realistic behavior. 3.4 Evaluation outcomes are recorded and analyzed to ensure garment performance.
4. Collaborate with technical design teams for refinement	4.1 Design prototypes and fit analysis reports are prepared and shared. 4.2 Feedback from the technical team is documented and integrated.

	4.3 Suggested pattern modifications and garment construction adjustments are discussed and prioritized. 4.4 Technical specifications, including tolerances and construction details, are aligned. 4.5 All refinement activities are tracked and archived.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Fit issues	1.1. Pattern Translation Errors 1.2. Improper Grading and Sizing 1.3. Particle distance

Curricular Content Guide

1. Underpinning Knowledge	1.1 Virtual human models (3D avatars) 1.2 Software features used to simulate fabric behavior and garment fit in a 3D environment. 1.3 Problems identified during fitting sessions 1.4 Insights and comments gathered after testing the garment fit on digital avatars. 1.5 Fit analysis results 1.6 Changes made to the original pattern based on fit analysis 1.7 The updated patterns after modifications 1.8 Simulation of the avatar's movements 1.9 The results of the fit analysis 1.10 Design prototypes and fit analysis reports
2. Underpinning Skills	2.1 Conducting virtual fit sessions using 3D avatars 2.2 Modifying patterns based on fit analysis 2.3 Evaluating garment comfort and movement 2.4 Collaborating with technical design teams for refinement
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 3D software 4.4 Lighting

	4.5 3D garment models
	4.6 Accessibility to the workplace

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 conducted virtual fit sessions using avatars 1.2 modified patterns based on fit analysis 1.3 evaluated garment comfort and movement 1.4 collaborated with technical design teams for refinement
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: COLLABORATE IN DIGITAL PRODUCT DEVELOPMENT CYCLE	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG-3FD-05-O
Unit Descriptor: This unit covers the skills, knowledge and attitudes required to collaborate in digital product development cycle. It specifically includes the task of collaborating with teams using 3D design tools, incorporating feedback to optimize virtual samples and optimizing approval processes with 3D tools.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Collaborate with teams using 3D design tools	1.1 <u>Colorway prototype</u> is created and shared with relevant teams. 1.2 Feedback from design, merchandising, and production teams are collected and reviewed. 1.3 <u>Visual presentations and virtual samples</u> are generated using 3D tools. 1.4 Design intent and construction details are communicated. 1.5 Real-time review sessions are conducted. 1.6 <u>Iterative changes</u> are applied and re-simulated to validate improvements. 1.7 Engagement records, comments, and version histories are documented.
2. Incorporate feedback to optimize virtual samples	2.1. Feedback from internal and external stakeholders are collected. 2.2. <u>Identified issues</u> are categorized and prioritized. 2.3. Virtual samples are updated with design and production requirements. 2.4. Modifications to patterns, textures, and simulation settings are applied. 2.5. Revised virtual samples are re-simulated and evaluated. 2.6. Before final approval, updated samples are shared. 2.7. All feedback loops and revisions are documented.
3. Optimize approval processes with 3D tools	3.1 3D garment samples are developed. 3.2 Approval-ready virtual samples are rendered. 3.3 Centralized platforms are used. 3.4 Stakeholder feedback is integrated. 3.5 Automated tools and digital annotation features are utilized. 3.6 Approved virtual samples are archived with version control. 3.7 Performance metrics time-to-approval and sample

	reduction rates are monitored.
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Range of Variables

Variable	Range (Includes but not limited to):
1. Colorway prototype	1.1 Color schemes 1.2 Platters 1.3 Materials finishes 1.4 Desing mockups
2. Visual presentations and virtual samples	2.1 3D software (CAD, CLO) 2.2 Visual output formats (Rendered images, VR samples) 2.3 Tools and technology (Animation, simulations)
3. Iterative changes	3.1 Minor adjustments 3.2 Major redesigns 3.3 Simulation 3.4 Physical testing 3.5 Prototype phases
4. Identified issues	4.1 Technical 4.2 Aesthetic 4.3 Functional 4.4 Market-driven

Curricular Content Guide

1. Underpinning Knowledge	1.1. The process of simulating garment movement and behavior in a 3D environment. 1.2. Different types of reports 1.3. The process of displaying garments in a digital environment 1.4. Virtual representations of garments 1.5. The process of reviewing digital garment models
2. Underpinning Skills	2.1 Creating garments animation 2.2 Generating different types report 2.3 Preparing and presenting visualizations 2.4 Integrating digital garment models 2.5 Incorporating digital garment models
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	4.1 Workplace (simulated or actual)

	4.2 Different types of report 4.3 3D software 4.4 Computer/ laptop 4.5 Accessibility to the workplace
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 collaborated with teams using 3D design tools 1.2 incorporated feedback to optimize virtual samples 1.3 optimized approval processes with 3D tools
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	3D Fashion Design
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S. N.	Major Equipment and Training facilities	Required facilities
1.	Computers/Laptop (Minimum Core i5, 16GB RAM, 500GB SSD, 21" Monitor, Graphics-supported)	20
2.	Software: CLO or BROWZWEAR	20
3.	Portable scanner (legal size)	01
4.	UPS/Backup Power Supply (1–2 hours backup per workstation)	20
5.	Projector & Screen (HD quality, HDMI support, 3000+ lumens)	1
6.	Mannequins/Dress Forms (Female and male standard sizes)	2

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on 3D Fashion Design, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.